



SUMMER INTERNSHIP REPORT

A INTERNSHIP REPORT

Submitted by

G.SANJAY PRATHAP (812022114033)

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

DEPARTMENT MECHANICAL ENGINEERING

(Internship Duration: Start Date to End Date)

(05.07.2025 to 05.08.2025)

M.A.M COLLEGE OF ENGINEERING AND TECHNOLOGY

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JULY 2025

ANNA UNIVERSITY: CHENNAI
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BONAFIDE CERTIFICATE

This is to certify that, **G.SANJAY PRATHAP** has satisfactorily completed the 4 Week Summer internship at “ **LS AUTOMOTIVE INDIA PRIVATE LIMITED , Ponnammallee - Thiruvallur High Road , Gudapa Ponnammallee Taluk , Thiruvallur District , Tamil Nadu** ”. This report is being submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in **Department of Mechanical Engineering** to **M.A.M College of Engineering and Technology**, under my guidance.

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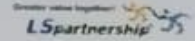
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Submitted for the Anna University Viva Voce Examination held on _

INTERNAL EXAMINER

EXTERNAL EXAMINER

CERTIFICATE



LS Automotive India Private Limited (formerly known as Daesung Electric India Pvt. Ltd.,)

LSAI/HRD/INT-021
30/07/2025

TO WHOM SO EVER IT MAY CONCERN

This is to certify that **Mr. G.SANJAY PRATHAP**, (Reg no: 812022114033) BE-MECH, a Student of M.A.M COLLEGE OF ENGINEERING AND TECHNOLOGY TRICHY has successfully completed an Internship in Production Department **05 JUL 2025 to 05 AUG 2025** under guidance of HOD, Production Department.

During the internship period, he actively participated in tasks related to:

- Shop floor observation and process familiarization
- Work instructions and production flow documentation
- Quality checks and basic 5S awareness
- Support in day-to-day production tracking and reporting
- Exposure to machinery operation under supervision

His performance was evaluated as Excellent and he displayed keen interest, discipline and a willingness to learn throughout the training.

We appreciate his contribution and wish him success in future professional pursuits.

For **LS Automotive India Pvt Ltd**


Authorised Signatory



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DECLARATION

I **G.SANJAY PRATHAP** hereby declare that the Internship Report submitted to the **M.A.M. College of Engineering and Technology** in partial fulfillment of the requirements for the award of the **Degree of Bachelor of Engineering** in **Department of Mechanical Engineering** is a record of original training undergone by me during the period **July 2025** under the supervision and guidance of **Mr.C.SASIDHARAN**, Assistant Professor, Department of Mechanical Engineering, M.A.M. College of Engineering and Technology and it has not formed the basis for the award of any Degree/Fellowship or other similar title to any candidate of any University.

Place:

Signature of the Student

Date:

ACKNOWLEDGEMENT

With warm hearts and immense pleasure, I thank the almighty for his grace and blessing bestowed on me, which drove me to the successful completion of this project. I take this opportunity to express my sincere thanks to the respected Director **Dr. M. A. Maluk Mohammed, M.E., Ph.D.**, and Secretary & Correspondent **Mrs. Fathima Bathool Maluk** who is guiding light for all activities in our college.

I express my sincere and humble tone of thanks to our Principal **Dr. X. Susan Christina, M.E., Ph.D.**, for providing me with all facilities needed for the successful completion of my work.

I would like thank to our Head of the department **Dr.S.Roseline, M.E.,Ph.D.**, for her cooperation, advice and suggestions at every stage of my project work.

I am very proud to extend my sincere thanks and gratitude to our Supervisor **Mr.C.Sasidharan,M.E.**, Assistant Professor, Department of Mechanical Engineering, M.A.M. College of Engineering & Technology, for his excellent guidance, advice and encouragement which boosted up our energy throughout the project Development.

I also thank all the teaching staff and non-teaching staff of the Department of Mechanical Engineering, my parents, and all my friends for their help and support to complete this project successfully.

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CHAPTER I

Company Introduction



1.1 Overview of LS Automotive India Private Limited

LS Automotive India Private Limited is a prominent manufacturer of automotive electrical components and a subsidiary of LS Automotive Technologies, headquartered in South Korea. The company has established its Indian operations at **Poonamallee, Chennai**, Tamil Nadu to support the rapidly growing automotive sector in India.

LS Automotive India specializes in designing and manufacturing advanced electrical systems for vehicles, with a strong emphasis on quality, innovation, and cost-effective solutions. The facility at Poonamallee is equipped with modern machinery, automated production lines, and rigorous quality control processes that meet global industry standards.

1.2 Global Presence and Parent Company

LS Automotive India operates under **LS Automotive Technologies Co., Ltd.**, a global leader in automotive electrical systems and a key member of the **LS Group**, a major South Korean industrial conglomerate.

With a global presence that includes manufacturing and R&D centers in **South Korea, China, India, Vietnam, Poland, the United States, and Brazil**, LS Automotive is strategically positioned to serve major global automotive markets. The company's focus on continuous R&D and digital transformation ensures it remains at the forefront of automotive electrification and smart mobility solutions.

1.3 Major Clients

LS Automotive India supplies components to many top automobile manufacturers (OEMs) in India and abroad. Some of the major clients include:

Hyundai Motor India Ltd.

Kia Motors India

Mahindra & Mahindra

Tata Motors

Maruti Suzuki

Renault-Nissan Alliance

Volkswagen Group (select markets)

These partnerships reflect the company's strong reputation for delivering reliable and high-performance automotive components.

1.4 Key Products

LS Automotive India manufactures a wide range of high-quality electrical and electronic components used in modern vehicles. The major product categories include:

Wiring Harnesses: Complex cable assemblies that distribute power and signals across various vehicle systems.

Relays: Electromechanical devices used to control high-current circuits with low-power signals.

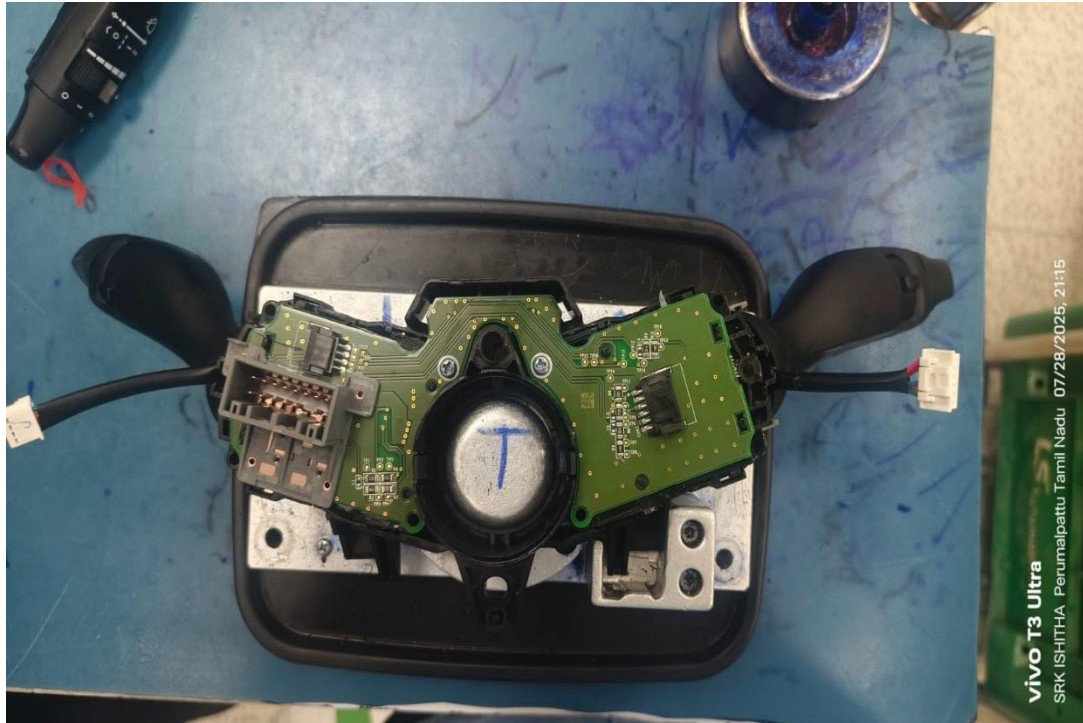
Switches: Includes ignition switches, power window switches, combination switches (lights, wipers), and other control switches.

Sensors: Used for detecting speed, temperature, pressure, and proximity in automotive systems.

Each of these components plays a critical role in ensuring vehicle safety, comfort, and functionality. With a commitment to innovation and customer satisfaction, LS Automotive India continues to be a key player in the electrical automotive components sector.

CHAPTER II

Overview of Automotive Components



2.1 Types of Automotive Components

Automotive components are broadly categorized into two main types:

a. Mechanical Components

These are physical parts involved in the movement, control, and structure of a vehicle. Common examples include:

Brake Pads and Discs – Help stop the vehicle by converting kinetic energy into heat through friction.

Gears and Transmissions – Transfer power from the engine to the wheels with different speed and torque ratios.

Suspension and Shock Absorbers – Maintain ride comfort and vehicle stability.

Clutch and Flywheel – Engage and disengage the engine from the transmission system.

b. Electrical and Electronic Components

These parts manage power distribution, sensing, and control functions:

Switches – Such as power window switches, ignition switches, headlamp and wiper switches.

Fuses and Relays – Protect electrical circuits and control power flow.

Sensors – Detect temperature, speed, pressure, and provide data to the ECU.

Wiring Harnesses – Carry electrical signals and power to various components.

2.2 Importance in Vehicle Performance and Safety

Automotive components play a vital role in ensuring the vehicle operates efficiently, safely, and reliably.

Safety: Components such as **brake systems, airbags, sensors, and anti-lock braking systems (ABS)** directly impact driver and passenger safety.

Performance: Proper functioning of **engine parts, transmission, and suspension** ensures smooth acceleration, gear shifting, and handling.

Efficiency: Well-integrated **electronic control systems** optimize fuel usage, emissions, and power output.

Comfort and Convenience: Devices like **climate control units, infotainment systems, and automated window controls** enhance the user experience.

2.3 Trends in Automotive Electrification

The automotive industry is undergoing a transformation with the rise of electric vehicles (EVs) and advanced driver-assistance systems (ADAS). Key trends include:

Increased Use of Electronics: Modern vehicles have more ECUs (Electronic Control Units), sensors, and software integration than ever before.

Shift to Electric Powertrains: Traditional engines are being replaced with electric motors, requiring new components such as inverters, battery management systems, and high-voltage connectors.

Lightweight Materials: To improve fuel economy and range, manufacturers are using **lightweight alloys and composites** in both mechanical and electrical parts.

Smart Components: Switches, sensors, and modules are becoming intelligent, capable of real-time diagnostics and communication via CAN or LIN networks.

Integration with IoT and Telematics: Components now support connectivity features for real-time vehicle monitoring, predictive maintenance, and enhanced navigation.

CHAPTER III

Electrical Components in Vehicles



In modern vehicles, electrical components play an essential role in ensuring the functionality, safety, efficiency, and comfort of the automobile. From basic lighting systems to advanced engine management and infotainment systems, these components form the nervous system of today's vehicles.

3.1 Switches

Switches are user-operated devices that control electrical circuits within the vehicle. They enable or disable specific functions and are essential for both comfort and operation.

Power Window Switches: These are mounted on the door panels and allow passengers to raise or lower windows. They use a momentary contact mechanism that activates window motors.

Ignition Switch: This is a crucial switch located on the steering column or dashboard. It initiates the engine start-up process and controls power to major systems like the ECU, fuel pump, and ignition coil. In modern vehicles, it may be replaced with a **start/stop button** integrated with keyless entry.

3.2 Relays and Fuses

Relays

Relays are electrically operated switches that control high-current circuits using a low-current signal from the ECU or control switch.

Used in systems like **fuel pumps, headlamps, wipers, and horn circuits**.

Provide electrical isolation between the control and power circuits, improving safety and reliability.

Fuses

Fuses protect vehicle wiring and components from short circuits and overloads.

They are placed in **fuse boxes** located in the engine bay and dashboard.

When excessive current flows, the fuse melts and breaks the circuit, preventing fire or equipment damage.

3.3 Sensors

Sensors convert physical parameters (like temperature, pressure, or motion) into electrical signals that the vehicle's control units can interpret.

Temperature Sensors: Used in **engine coolant, intake air, and cabin temperature** monitoring. Help optimize combustion, protect engine from overheating, and regulate climate control systems.

Speed Sensors: Detect the rotational speed of the wheels or transmission. Essential for **ABS, speedometers, and traction control systems**.

Pressure Sensors: Monitor **fuel pressure, tire pressure (TPMS), oil pressure**, and more. They are crucial for both performance and safety functions.

Sensors ensure that the vehicle reacts intelligently and efficiently to varying operating conditions.

3.4 Control Units (ECM Basics)

The **Electronic Control Module (ECM)**, also known as the **Engine Control Unit (ECU)**, is the brain of the vehicle's engine management system.

It processes inputs from sensors (like air flow, throttle position, and oxygen sensors) to manage outputs such as fuel injection, ignition timing, and emissions control.

Uses embedded software and microcontrollers to perform real-time calculations and decision-making.

Communicates with other modules like the **Transmission Control Unit (TCU)**, **Body Control Module (BCM)**, and **ABS module** using vehicle networks such as **CAN (Controller Area Network)**.

As vehicles evolve into smart and connected machines, the number and complexity of control units continue to increase.

CHAPTER IV

Manufacturing Process of Automotive Components

The production of automotive electrical and mechanical components involves a series of precise, high-quality manufacturing processes. These processes are designed to ensure that every component meets stringent performance, safety, and durability standards. LS Automotive India employs both manual and semi-automated systems to enhance efficiency and maintain product consistency.

4.1 Stamping and Molding

Stamping is a process where flat metal sheets (usually copper, brass, or steel) are shaped into desired forms using a press machine and dies. This is commonly used for:

- Terminal connectors

- Relay contact plate

- Switch contact points

The high-speed precision of stamping ensures consistent part dimensions, which is critical for electrical conductivity and mechanical fit.

Molding refers to forming materials like thermosetting plastics into a defined shape. It is widely used in:

- Switch housings

- Relay bases

- Sensor enclosures

Molding helps achieve complex geometries and insulative properties essential for electrical safety.

4.2 Soldering and Crimping

Soldering is the process of joining electrical contacts and wires using a filler metal (solder), typically done in printed circuit boards (PCBs) and relay modules. It provides:

Strong electrical connectivity

Resistance to vibration and thermal stress

Crimping involves mechanically joining wires to terminals without solder, using a crimping tool or press. This is commonly used in:

Wiring harness assemblies

Connector manufacturing

Crimping is faster than soldering and offers high mechanical strength and conductivity when done properly.

4.3 Injection Molding for Plastic Parts

Injection molding is used to create plastic components with precise dimensions and surface finishes. Granulated plastic material is melted, injected into a mold, cooled, and ejected.

Used for manufacturing:

Switch covers and buttons

Connector housings

Sensor casings

Relay covers

Advantages of injection molding:

High production rate

Consistency and repeatability

Complex shapes with tight tolerances

This process plays a vital role in enclosing and protecting electrical circuits from dust, moisture, and heat.

4.4 Assembly Lines: Manual and Semi-Automated

At LS Automotive India, the final product assembly is performed on structured **manual and semi-automated assembly lines**. These lines integrate different parts and sub-assemblies into a complete functional unit.

Manual Assembly

Workers manually insert and fit parts using jigs, fixtures, and basic hand tools.

Common in small-volume or variable-configuration products.

Allows real-time visual inspection and quality checks.



Semi-Automated Assembly

Machines handle repetitive or precise operations like screw tightening, connector insertion, or testing.

Human operators perform supervision, part feeding, and final adjustments.

Increases consistency, reduces operator fatigue, and improves cycle time.

Both systems are supported by **quality checkpoints**, where products are visually and electrically tested before moving to the next stage or packaging.

CHAPTER V

PCB Assembly and Quality Testing

Printed Circuit Boards (PCBs) are the foundation of modern automotive electronics, housing and interconnecting critical components such as microcontrollers, sensors, and relays. In LS Automotive India, precise PCB assembly and rigorous quality testing are essential to ensure performance, durability, and safety of automotive electrical systems.

5.1 Surface Mount Technology (SMT)

Surface Mount Technology (SMT) is a method for producing electronic circuits where components are mounted directly onto the surface of PCBs.

SMT Process Steps:

Solder Paste Printing: A stencil is used to apply solder paste on specific areas of the PCB where components will be placed.

Pick and Place: High-speed machines place tiny surface-mount components (resistors, ICs, capacitors) accurately on the PCB.

Reflow Soldering: The assembled boards are passed through a reflow oven where the solder paste melts and solidifies, securing the components to the board.

Advantages:

Higher component density

Faster assembly process

Smaller and more reliable electronics

SMT is used extensively in automotive applications for compact and vibration-resistant electronic modules.

5.2 Automated Optical Inspection (AOI)

AOI (Automated Optical Inspection) is a non-contact visual inspection technique used to detect assembly defects and quality issues on PCBs.

Key Functions of AOI:

Detects **misaligned components, solder bridges, missing parts, wrong polarity, and open joints**

Uses **high-resolution cameras and image-processing algorithms**

Performed **immediately after SMT** and before final testing to catch early-stage defects

AOI significantly reduces manual inspection time and improves accuracy in high-volume production environments like LS Automotive's SMT lines.

5.3 Functional Testing

Functional testing ensures that the PCB performs its intended operations under simulated or real working conditions.

Types of Functional Tests:

Power-On Testing: Verifies voltage levels, current draw, and initial signal output when power is applied to the circuit.

Relay Function Testing: Checks relay response, contact switching behavior, and current handling under load.

Switch and Button Response: Ensures mechanical switches on the PCB respond correctly to user inputs.

At LS Automotive India, each board undergoes a series of functional tests using test jigs and diagnostic software to ensure 100% quality before packaging.

5.4 Reliability Checks

Automotive electronics are exposed to harsh environments. To ensure durability, **reliability testing** is carried out on a sample batch or in final qualification stages.

Common Reliability Tests:

High-Temperature Testing: PCBs are exposed to elevated temperatures (typically 85°C or higher) to test thermal resistance.

Vibration Testing: Simulates the mechanical vibration stress experienced in vehicles during operation to check for solder or component failure.

Moisture and Humidity Testing: Evaluates corrosion resistance and insulation performance under high humidity or condensation.

These tests are vital to qualify PCBs for long-term automotive use, ensuring customer safety and reducing field failures.

CHAPTER VI

Wiring Harness and Assembly Line Operations

Wiring harnesses serve as the central nervous system of modern vehicles, connecting electrical and electronic components to deliver power and signals throughout the vehicle. LS Automotive India Pvt. Ltd. plays a significant role in designing, assembling, and testing high-quality wiring harnesses for major automobile manufacturers.

6.1 Harness Design and Cable Selection

The **wiring harness design** process involves planning the layout of wires and connectors to suit the specific electrical architecture of the vehicle. Key aspects include:

Circuit Mapping: Determining the number and type of circuits needed for systems like lighting, sensors, ignition, infotainment, etc.

Cable Selection: Based on current rating, voltage, flexibility, temperature tolerance, and insulation type (PVC, XLPE).

Connector Selection: Choosing suitable connectors (sealed or unsealed) based on location and environmental exposure.

Customization: Each vehicle model requires a unique harness layout, which is precisely engineered using CAD software.

The design ensures **space optimization**, **EMI protection**, and **reliability** under operating conditions.

6.2 Crimping, Routing, and Taping

After design, the harness goes through various **manual and semi-automated assembly processes**:

Crimping

Wires are terminated with metal terminals using crimping machines.

Crimping ensures strong mechanical bonding and low electrical resistance.

Visual inspection and pull-force testing are used to ensure crimp quality.

Routing

Wires are laid out on a **harness board (layout board)** based on design templates.

Technicians follow specific wire color codes and lengths to organize routing.

Taping and Protection

Tapes, sleeves, and tubes are applied to bind the wires and protect them from abrasion, moisture, and heat.

Specific taping patterns are followed to support branch junctions and flexibility at connectors.

These processes ensure **durability, serviceability, and protection** throughout the harness lifecycle.

6.3 Electrical Testing (Continuity & High Voltage)

Before dispatch, each wiring harness is subjected to rigorous electrical tests to ensure functionality and safety.



Continuity Testing

Verifies that current flows through the intended path without any open circuits.

Detects missing or misplaced terminals and faulty connections.

High Voltage (HiPot) Testing

Applies high voltage (typically 1000V or more) between circuits to check insulation resistance.

Identifies short circuits, poor insulation, or leakage paths.

Both tests are done using automated test benches with **pass/fail indicators** to ensure compliance with OEM standards.

6.4 Final Packing and Barcode Labelling

After inspection and testing, the wiring harness is prepared for shipment:

Final Packing: Harnesses are bundled, labeled, and packed in cartons or racks to prevent deformation during transportation.

Barcode Labelling: Each harness is tagged with a **barcode label** that includes part number, model, production date, and traceability code.

Documentation: Packing lists and quality certificates are generated to support inventory and delivery tracking.

Proper packing and labelling ensure the right harness reaches the right vehicle assembly line in perfect condition.

CHAPTER VII

Day-wise Internship Activities

This section details my daily learning and experiences during the 29-day internship at **L S Automotive India Private Limited**, Poonamallee, Chennai. I was rotated through multiple departments, gaining exposure to various aspects of automotive and electrical component manufacturing including design, assembly, testing, quality control, and process documentation.



Day 1 – Orientation and Safety Briefing

Introduced to the company's organizational structure, products, and manufacturing layout.

Attended safety induction session covering PPE usage, emergency procedures, fire exits, and electrical hazard zones.

Understood the importance of the 5S methodology and workplace discipline in an automotive production environment.

Day 2–3 – Switch and Relay Production Process

Observed the manufacturing of **ignition switches**, **window switches**, and **combination switches**.

Learned the **relay assembly** process including coil winding, armature installation, and contact welding.

Assisted in simple relay sub-assembly and function check procedures using test benches.

Day 4–5 – PCB Assembly and Testing

Gained hands-on exposure to **SMT line operations**, including solder paste printing, pick-and-place, and reflow oven.

Conducted visual inspections and assisted in **AOI (Automated Optical Inspection)** review.

Participated in **Power-On Testing** and **relay function tests** for PCB-based modules.

Day 6–7 – Wiring Harness Production: Crimping & Routing

Learned **wire cutting, crimping, and stripping** techniques using semi-automated machines.

Practiced routing wires on layout boards based on assembly drawings.

Understood the importance of branch taping, terminal pin checks, and connector locking.

Day 8–10 – Wiring Harness: Bundling, Taping & Sub-Assembly

Assisted in bundling harnesses using tape and sleeves.

Applied anti-vibration tapes and insulation wraps.

Participated in pre-inspection of completed sub-harness assemblies for missing pins and open circuits.

Day 11–13 – Electrical Testing of Harnesses

Performed **continuity testing** using automated testers to detect open or crossed wires.

Assisted in **High Voltage (HiPot) testing** for insulation verification.

Helped document pass/fail test results and created batch inspection logs.

Day 14–16 – Switch and Connector Quality Checks

Worked with the QC team to inspect:

Switch click force and travel consistency.

Relay contact bounce and coil resistance.

Connector fit and locking accuracy.

Participated in defect logging and learned about Poka-Yoke (error-proofing) techniques.

Day 17–18 – Assembly Line Operations: Manual Stations

Supported operators in manual assembly of switch modules.

Learned how jigs and fixtures are used for alignment and efficiency.

Assisted in visual inspections and basic mechanical function checks.

Day 19–20 – Assembly Line Operations: Semi-Automated Stations

Observed machine-controlled torque tightening, ultrasonic welding, and automated tapping.

Gained experience in reading SOPs and work instructions.

Practiced barcode scanning and product tracking on the line.

Day 21–23 – Final Product Testing & Visual Inspection

Inspected final assemblies for cosmetic and functional quality.

Performed **tactile feel tests** for switches and **mechanical actuation tests**.

Learned about IP rating tests for dust and water resistance in select components.

Day 24–25 – Packing and Labelling

Participated in **final packing** of relays, switches, and wiring harnesses.

Assisted in applying **barcode labels**, product tags, and part numbers.

Helped prepare shipping documentation and inventory records.

Day 26–27 – Documentation and Audit Support

Helped maintain **process audit sheets**, operator logbooks, and production reports.

Supported supervisors in tracking **first pass yield (FPY)** and **defect per million (DPM)** data.

Understood the significance of traceability and batch-wise documentation.

Day 28–29 – Report Compilation, Team Feedback & Exit Discussion

Compiled and organized my report and personal logbook for submission.

Participated in team discussions about process improvements and safety suggestions.

Attended a closing feedback session with the HR and department head; received appreciation for punctuality and involvement.

CHAPTER VIII

Conclusion

My 29-day internship at **L S Automotive India Private Limited**, Poonamallee, was a transformative and enriching experience that provided me with deep insight into the manufacturing processes of automotive and electrical components. The opportunity to observe and actively participate in various departments — including switch assembly, relay production, PCB testing, wiring harness fabrication, and quality assurance — allowed me to apply my academic knowledge in a real-world industrial environment.

Key Learnings from LS Automotive

I gained a comprehensive understanding of how **automotive electrical components** are designed, assembled, tested, and delivered to global clients.

I observed **process flow**, from raw material handling and component assembly to testing, packing, and final dispatch.

I learned the importance of **standard operating procedures (SOPs)**, **process discipline**, and **lean manufacturing principles** such as 5S and Kaizen in achieving efficiency and quality.

Skills Developed

Technical Skills:

Operation and inspection of **relays, switches, sensors, and wiring harnesses**

Exposure to **PCB assembly** techniques including SMT and AOI systems

Knowledge of **electrical testing** tools such as continuity testers, HiPot testers, and functional bench

Hands-on practice in **crimping, routing, taping, and final visual inspections**

Professional Skills:

Teamwork and collaboration with technicians, supervisors, and fellow interns on shop floor activities

Improved **communication skills** through report logging, technical discussions, and feedback sessions

Learned to maintain **discipline, punctuality, and professionalism** in a structured industrial setting

Real-World Exposure

This internship gave me valuable exposure to **large-scale automotive component manufacturing systems**. I understood how precision, consistency, and traceability are maintained across thousands of parts produced each day. Interacting with real-time production data, quality control documentation, and live assembly lines helped me bridge the gap between theory and practical engineering.

Overall, the internship has strengthened my foundation in automotive manufacturing and motivated me to pursue further learning in the areas of **automotive electronics, quality systems, and industrial automation**.

CHAPTER IX

Impact Analysis

The internship at **L S Automotive India Private Limited** has had a significant impact on both my academic understanding and professional development. It provided a real-world perspective on the end-to-end lifecycle of automotive and electrical components — from procurement and production to inspection and delivery — and helped me connect theoretical concepts from my mechanical engineering curriculum to actual industrial applications.

Understanding Supply Chain and Production Lifecycle

During the internship, I gained valuable insight into the **automotive component supply chain**, which includes:

Procurement of raw materials and electronic parts

Batch-wise production planning

Quality control at each stage

Inventory management and delivery tracking

I observed how different departments such as **production, quality assurance, logistics, and documentation** are integrated to ensure timely and efficient product output. This understanding of the **production lifecycle** helped me appreciate how large-scale manufacturing units maintain consistency, safety, and performance in a competitive automotive market.

Improved Technical Knowledge

My technical understanding was enhanced in the following areas:

Sensors: Learned about different types used in vehicles (temperature, speed, pressure) and their testing protocols.

Switches and Relays: Gained hands-on experience in their assembly, testing, and quality inspection.

Electronic Control Units (ECUs): Understood their role in managing functions like ignition, lighting, and safety systems, and observed PCB-level control modules in production.

This technical exposure will be highly beneficial in future projects and labs related to **mechatronics, electronics, and vehicle dynamics**.

Alignment with Mechanical Engineering Curriculum

The internship directly complemented subjects I have studied such as:

Manufacturing Technology – Assembly line procedures, material handling, and production tools

Electrical and Electronics Engineering – Circuit basics, testing, and relay operations

Quality Control and Maintenance – Defect analysis, inspection techniques, documentation

Mechatronics and Automation – Semi-automated stations, sensor-based systems, process control

It allowed me to observe how academic theories are implemented practically on the shop floor with high precision and efficiency.

Relevance to Future Career in Automotive and Mechatronics

This internship has reinforced my interest in pursuing a career in the **automotive and mechatronics domain**. The skills and exposure I received will help in:

Entering roles related to **automotive electronics, manufacturing systems, and quality engineering**

Preparing for higher studies or certification in **embedded systems, vehicle electrification, or industrial automation**

Adapting quickly to roles in **R&D, design, production, or testing** in companies serving the automotive sector

In conclusion, the internship has not only expanded my technical skill set but also provided clarity on career direction and personal goals in the field of modern automotive engineering.